

1	2	3	4	5	6	Optional	Sum
/20	/20	/20	/20	/20	/20	/20	/100

Name _____

Dec. 24, 2003 Wednesday

Department _____

SID _____

University Physics

Sample Problems for Final Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

Electron charge magnitude	$e = 1.60 \times 10^{-19} \text{ C}$
Dielectric constant of vacuum	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
Permeability of vacuum	$\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$
Vacuum speed of light	$c = 3.00 \times 10^8 \text{ m/s}$
Electron rest mass	$m_e = 9.11 \times 10^{-31} \text{ kg} = 0.511 \text{ MeV}/c^2$
Proton rest mass	$m_p = 1.672 \times 10^{-27} \text{ kg} = 938.3 \text{ MeV}/c^2$
Neutron rest mass	$m_n = 1.675 \times 10^{-27} \text{ kg} = 939.6 \text{ MeV}/c^2$
Atomic mass unit	$1 \text{ amu} = 1.660 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$
Electron volt (eV)	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
Planck constant	$h = 6.63 \times 10^{-34} \text{ Js}$ $\hbar = 2.11 \times 10^{-34} \text{ Js} = 0.66 \times 10^{-15} \text{ eVs}$
Boltzmann constant	$k_B = 1.38 \times 10^{-23} \text{ J/K} = 8.62 \times 10^{-5} \text{ eV/K}$
Bohr radius	$a_0 = 0.259 \text{ Angstroms}$

Select five out of following six problems.

- (20pts). (a) In Compton scattering, find the maximum increase of the wavelength of the photon.
(b) If the wavelength of the scattered photon is the twice of the wavelength of the incident photon, what is the minimum energy of the incident X-ray?

- (20 pts) (a) Write the time-dependent Schrödinger equation and the stationary (time independent) Schrödinger equation for a particle of mass m moving in the one-dimensional harmonic potential

$$U(x) = \frac{1}{2} kx^2$$

(b) The solution for the stationary Schrödinger equation in the one-dimensional harmonic potential is

$$E_n = (n + \frac{1}{2})\hbar\omega, \quad \psi_n(x) = N_n e^{-\frac{\alpha^2 x^2}{2}} H_n(\alpha x), \quad n = 1, 2, 3, \dots$$

where $\omega = \sqrt{k/m}$, $\alpha = \sqrt{\mu\omega/\hbar}$, $N_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{\frac{1}{2}}$ and $H_n(\xi)$ is the Hermite polynomial

with the highest power n . We know that $H_n(\xi)$ is an even or odd function if n is even or odd. If there is a particle of mass m moving in the potential

$$U(x) = \begin{cases} \frac{1}{2} kx^2, & x \geq 0, \\ +\infty, & x < 0, \end{cases}$$

find the stationary wave functions and the energy levels of the particle.

3.(20pts) If we put 6 identical bosons or 6 identical spin-1/2 fermions into a one-dimensional infinite well of width a (the energy levels for a particle of mass m in the well is

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}).$$

(a) Draw a schematic diagram to show how the energy levels are filled in each case when the system is in the ground state.

(b) Find the ground state energy of the system in each case.

(c) The atomic mass of a copper atom is 63.55; the density is 8.96 gcm^{-3} . Calculate the Fermi energy E_F of the electron gas in copper. (Assume there is 1 valence electron per atom for copper.)

4. (20pts) Radioactive nuclei with decay constant λ are produced in an accelerator at a constant rate R_p .

(a) Derive the differential equation for the number of radioactive nuclei N

(b) If the number of the nuclei is zero at $t = 0$, find the number N as a function of time t .

(c) ^{62}Cu is produced at a rate 100 per second by placing ordinary copper(^{63}Cu) in a beam of high-energy protons and ^{62}Cu decays by beta-decay with a half-life of 10 min. How many ^{62}Cu are there after the time long enough so that $dN/dt \rightarrow 0$?

5.(20pts) The empirical binding energy of a nucleus can be written as

$$E_b = a_1 A - a_2 A^{2/3} - a_3 Z^2 A^{-1/3} - a_4 (N - Z)^2 A^{-1}$$

(the pairing energy is not included), where Z is the atomic number, A is the mass number and

N is the neutron number of the nucleus. The values of the constants are

$$a_1 = 15.8 \text{ MeV}, a_2 = 17.8 \text{ MeV}, a_3 = 0.71 \text{ MeV}, a_4 = 23.7 \text{ MeV}.$$

(a) Determine the atomic number of the most stable nucleus for a given mass number A .

(b) Explain the physics meaning of the result.

6. (20pts) Determine if the following processes are mediated mainly by strong, electromagnetic or weak interactions or forbidden. Explain why.

(a) $\eta \rightarrow \pi^+ + \pi^- + \pi^0$.

(b) $\rho \rightarrow \pi^+ + \pi^-$

(c) $\pi^+ \rightarrow \mu^+ + \nu_e$

(d) $K^0 \rightarrow \pi^+ + \pi^-$

not required !!!

Optional Problem (20pts) Consider a spinless positive pion decaying at rest. (The initial angular momentum and linear momentum of the pion are zero.) A positive muon and a muon neutrino are emitted in opposite directions with equal and opposite momenta. Meanwhile, the two emitted particles have equal but opposite spins due to the angular momentum conservation.

(a) Draw a figure to show the process and the mirror image process of the decay.

(b) Can the mirror image process occur in nature? Why?

(c) What interaction governs the positive pion decay?

(d) What conclusion can you draw if the mirror process can/cannot occur in nature?